

Supplementary Figures

Comparative Structural Modeling Suggest Divergent Conformational Plasticity and Surface Physicochemistry in Phytoene Synthase and Dehydrosqualene Synthase

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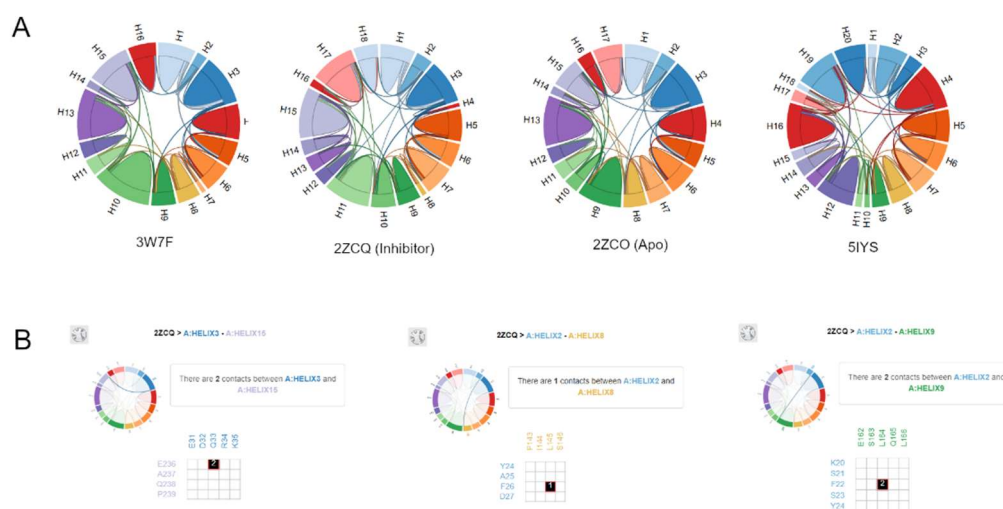


Figure S1 Helix Interaction Dynamics. (A) Chord diagrams representing inter-helix contacts for CrtM structures: Holo (3W7F), Inhibitor-bound (2ZCQ), Apo (2ZCO), and Holo (5IYS). The thickness of chords represents contact density; the Apo form exhibits a notable reduction in connectivity involving Helix-2 compared to ligand-bound states. (B) Detailed contact matrices for the 2ZCQ structure highlighting specific stabilizing interactions: Helix-3(Q33)–Helix-15(E236), Helix-2(F26)–Helix-8(L145), and Helix-2(F22)–Helix-9(L164). These interactions lock the binding cavity in the active conformation.

AMBER Trajectory Convergence Analysis (CES)

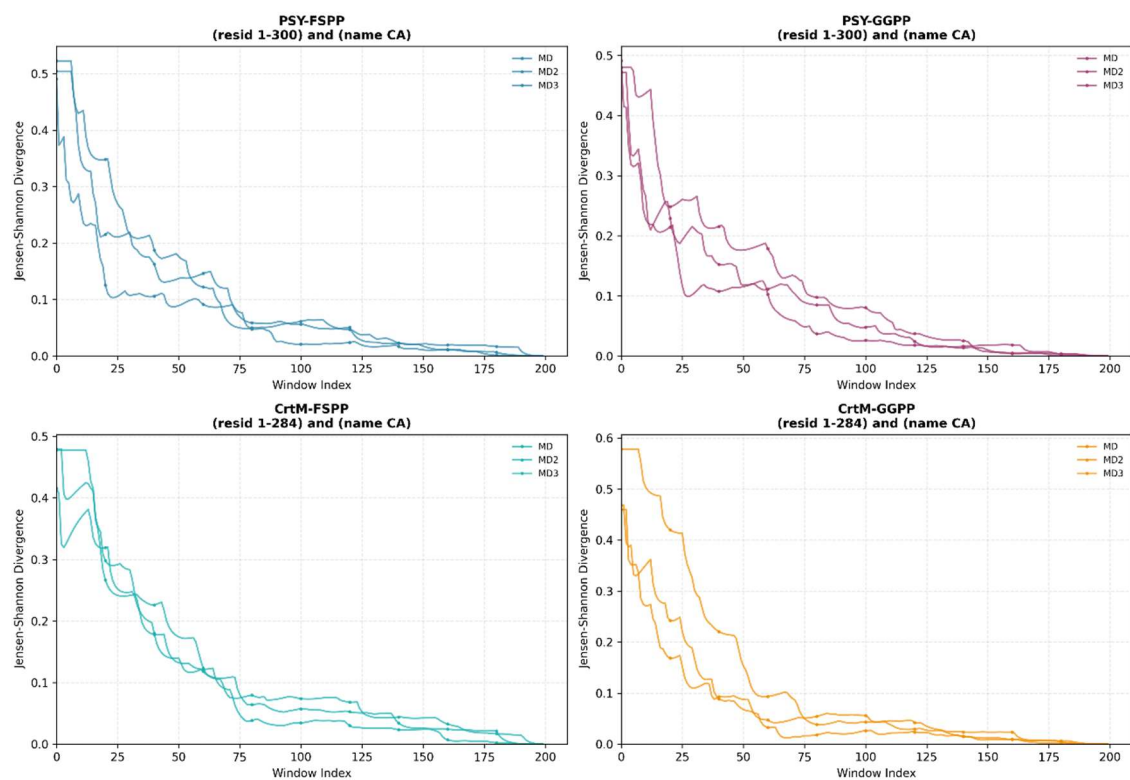


Figure S2. Convergence Analysis shows a plateau in the plot occurring between window times 50 and 75.

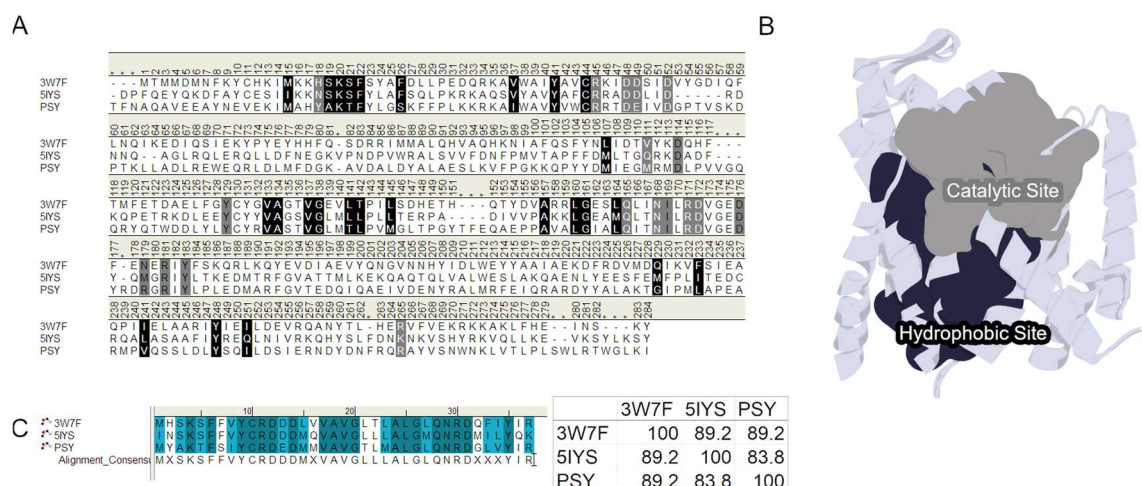
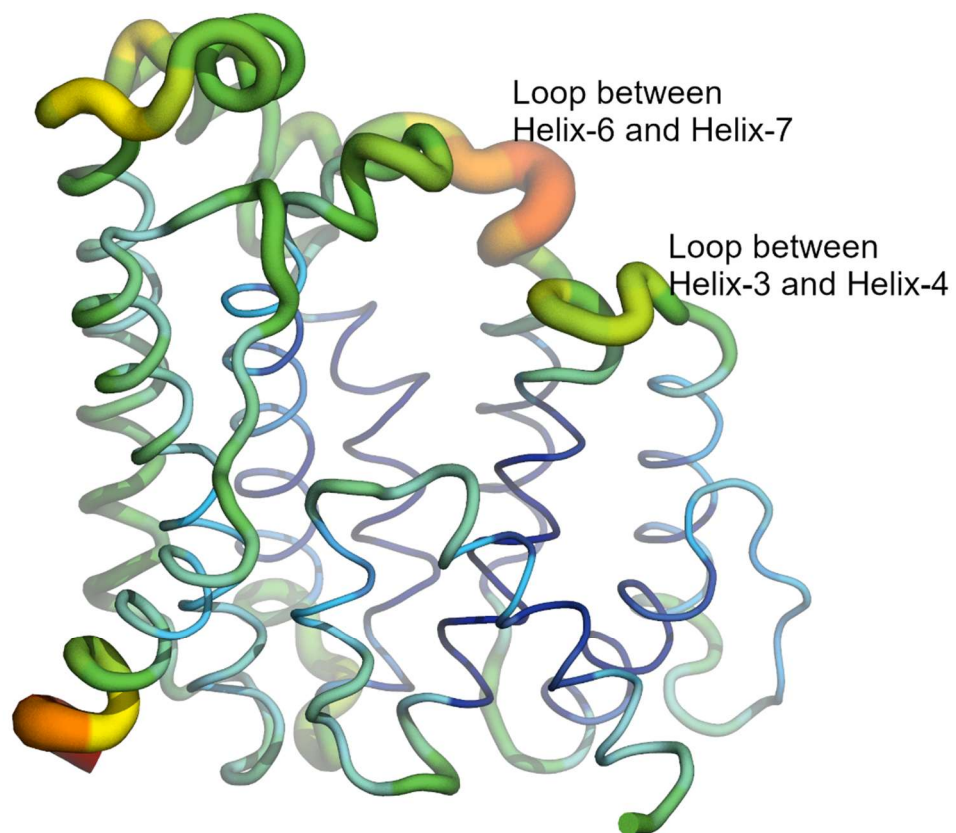


Figure S3. (A) Sequence alignment of CrtM (3W7F and 5IYS) and PSY (CrtB from *Euglena gracilis* without the irregular region sequence, accession No BAU24764.1). Residues corresponding to the ligand-binding site are highlighted, with catalytic residues shaded in gray and the hydrophobic site marked in black. (B) Structural presentation of the Catalytic Site (gray clouded area) and Hydrophobic Site (black clouded area). (C) The Binding cavity sequences similarity percentage.



CrtM (3W7F) b-factor

Figure S4. The B-factor value visualization of the 3W7F structure shows that the thicker the ribbon, the higher the B-factor value.

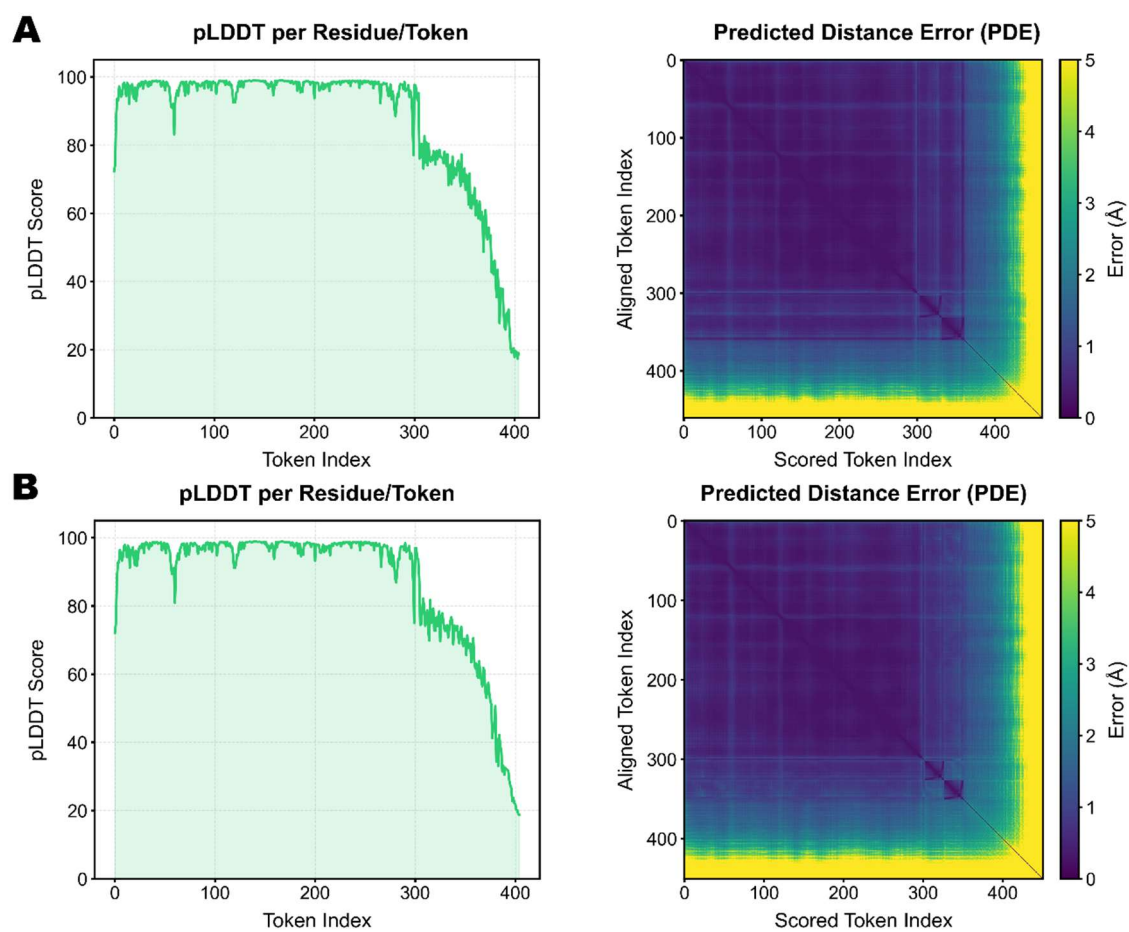


Figure S5. OpenFold prediction confidence metrics for PSY substrate complexes.

Table S1. Simulation Statistics

System	Replica	Metric	Mean	Std	Min	Max	Median	CV (%)
CrtM-FSPP	MD	RMSD	1.00	0.09	0.73	1.35	1.00	8.52
CrtM-FSPP	MD2	RMSD	1.13	0.11	0.82	1.58	1.12	9.34
CrtM-FSPP	MD3	RMSD	1.18	0.15	0.81	1.66	1.18	12.63
CrtM-FSPP	MD	Rg	19.69	0.05	19.49	19.91	19.69	0.27
CrtM-FSPP	MD2	Rg	19.83	0.06	19.65	20.13	19.83	0.31
CrtM-FSPP	MD3	Rg	19.87	0.08	19.59	20.12	19.88	0.40
CrtM-GGPP	MD	RMSD	0.92	0.11	0.63	1.37	0.91	12.05
CrtM-GGPP	MD2	RMSD	1.06	0.09	0.83	1.47	1.05	8.02
CrtM-GGPP	MD3	RMSD	1.05	0.10	0.73	1.44	1.04	9.72
CrtM-GGPP	MD	Rg	19.78	0.07	19.56	20.05	19.78	0.35
CrtM-GGPP	MD2	Rg	19.73	0.07	19.47	20.02	19.73	0.34
CrtM-GGPP	MD3	Rg	19.76	0.06	19.53	20.00	19.76	0.29
PSY-FSPP	MD	RMSD	1.25	0.18	0.82	2.14	1.23	14.69
PSY-FSPP	MD2	RMSD	1.30	0.24	0.83	2.09	1.25	18.13
PSY-FSPP	MD3	RMSD	1.35	0.21	0.84	2.11	1.33	15.33
PSY-FSPP	MD	Rg	19.85	0.07	19.62	20.10	19.85	0.33
PSY-FSPP	MD2	Rg	19.81	0.07	19.60	20.07	19.81	0.36
PSY-FSPP	MD3	Rg	19.88	0.12	19.57	20.32	19.86	0.59
PSY-GGPP	MD	RMSD	1.11	0.14	0.66	1.69	1.10	12.17
PSY-GGPP	MD2	RMSD	1.49	0.27	0.73	2.20	1.52	17.92
PSY-GGPP	MD3	RMSD	1.06	0.15	0.70	1.64	1.05	13.85
PSY-GGPP	MD	Rg	19.87	0.08	19.66	20.16	19.86	0.41
PSY-GGPP	MD2	Rg	19.97	0.07	19.74	20.31	19.97	0.33
PSY-GGPP	MD3	Rg	19.89	0.08	19.65	20.21	19.89	0.39